Smart Agricultural Technology 10 (2025) 100730

Contents lists available at ScienceDirect

Smart Agricultural Technology

journal homepage: www.journals.elsevier.com/smart-agricultural-technology

Smart UAV-assisted rose growth monitoring with improved YOLOv10 and Mamba restoration techniques

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ARTICLE INFO

Keywords:
Mamba
Precision agriculture
Rose growth monitoring
Super-resolution reconstruction
Unmanned aerial vehicle
YOLOv10

ABSTRACT

Recent advances in unmanned aerial vehicles (UAVs) technology and deep learning have revolutionized agricultural monitoring, yet challenges remain in processing low-resolution field imagery for precision floriculture. Here, we presented an innovative approach combining state-of-the-art super-resolution reconstruction (SRR) and object detection for accurate rose growth monitoring in large-scale greenhouse environments. We introduced MambaIR, a novel SRR algorithm based on selective state-space models, which significantly outperforms existing methods in enhancing low-resolution UAV imagery (PSNR: 28.34 dB, SSIM: 77.07 %). We also developed ROSE-YOLO, an improved object detection model tailored for rose identification, achieving 95.3 % mean average precision (mAP) on high-resolution images. The synergy between MambaIR and ROSE-YOLO enables 94.4 % mAP on reconstructed super-resolution images, nearly matching high-resolution performance. Through comprehensive experiments and Grad-CAM visualizations, we demonstrated our method's superior focus on key rose features and identify an optimal super-resolution magnification factor balancing detail enhancement and computational efficiency. This integrated approach overcomes resolution limitations in UAV-based agricultural monitoring, offering a scalable and accurate solution for rose growth assessment. Our method reduces technical barriers, offering a scalable and cost-effective solution for greenhouse monitoring by addressing low-resolution imagery challenges and enhancing decision-making processes. This research lays the groundwork for broader applications of UAV and AI technologies in sustainable agriculture. The findings pave the way for advanced, data-driven precision agriculture, integrating deep learning with remote sensing methodologies to improve floriculture management.

1. Introduction

By the end of 2023, the global floriculture industry had an annual output value exceeding $57 billion, driven primarily by increasing consumer demand for high-quality, aesthetically pleasing floral products [1]. However, the industry faces the challenge of optimizing production while minimizing environmental impact [2]. Roses, renowned for their unique beauty, fragrant aroma, and rich cultural symbolism, are among the most popular and commercially valuable cut flowers, making them a crucial component of this industry [3,4]. With the surge in demand, developing precise and efficient cultivation strategies has become essential to ensure that the quality and freshness of flowers meet market expectations [5,6]. This paper addresses these challenges by exploring innovative approaches to greenhouse rose monitoring, which enhance productivity while addressing labor and precision limitations.

Traditional methods for monitoring and managing rose cultivation, particularly in large-scale greenhouse environments, heavily rely on manual operations. These techniques are not only time-consuming and labor-intensive but are also prone to human error [7]. The need to monitor and record vast numbers of plants poses significant challenges in achieving the accuracy and efficiency required in modern agricultural practices. As the industry continues to expand, there is a growing need for more advanced, automated, and intelligent monitoring methods that minimize human intervention while providing real-time data and insights.

Recent advancements in unmanned aerial vehicles (UAVs) technology have opened new avenues for agricultural monitoring, offering the potential for rapid, large-scale data acquisition [8]. For instance, Liang

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https://doi.org/10.1016/j.atech.2024.100730
Received 7 September 2024; Received in revised form 4 December 2024; Accepted 15 December 2024
Available online 18 December 2024
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et al. [9] demonstrated the potential of these technologies in precision agriculture by using UAVs to collect data processed with advanced algorithms such as YOLOv5 and Transformers. Similarly, Shang et al. [10] successfully applied YOLO models to detect apple blossoms, showing that single-stage detection algorithms are well-suited for large, distinguishable flowers. However, challenges arise when detecting smaller or densely packed flowers [11]. Besides, applying UAVs in greenhouse floriculture presents unique challenges. Although UAVs can capture images over extensive areas, the resolution of these images often decreases with increasing flight altitude or distance, potentially leading to the loss of critical details [12]. This limitation is particularly pronounced in rose cultivation, where accurate growth assessment requires clearly identifying features such as the size and shape of individual flowers, which are crucial for detailed growth analysis [13]. Consequently, this limitation impedes the widespread application of UAV-based monitoring in precise rose growth monitoring.

Meanwhile, significant progress has been made in computer vision, particularly in super-resolution reconstruction (SRR) and object detection [14–16]. SRR technology aims to reconstruct high-resolution images from low-resolution inputs, thus alleviating issues related to insufficient resolution [17]. Previous SRR methods include interpolation, reconstruction, and machine learning algorithms, each with its limitations [18]. Nevertheless, deep learning-based SRR has shown substantial potential in image restoration. Recent advances, such as SRCNN, GANs, and Transformers, have propelled the field forward [19]. Notably, SRR models based on Mamba have recently introduced state-space modeling into image restoration, effectively preserving details and textures while maintaining computational efficiency and global receptive fields [20,21].

The combination of UAVs with SRR technologies has shown promising results in various applications, establishing a strong theoretical foundation for improving low-resolution imagery. For example, studies in structural monitoring [22] and road surface analysis [23] demonstrate the effectiveness of combining UAV imagery with SRR methods to enhance detail extraction and improve subsequent analyses. However, these applications predominantly focus on static targets, where environmental variables are less dynamic. In contrast, agricultural environments introduce unique challenges, including irregular object shapes, variable lighting conditions, and dynamic growth stages, requiring tailored SRR and object detection approaches.

Despite these advancements, the application of deep learning-based SRR in agricultural image processing remains limited. Recent efforts, such as Zhu et al. [24], have explored the integration of SRR and UAV technologies for specific agricultural tasks, such as disease detection in vineyards, illustrating the potential of these methods for dynamic environments. However, the effectiveness of SRR techniques in tasks like object detection and semantic segmentation in agricultural images has not been fully explored [25]. Additionally, the impact of various SRR networks and their scaling factors on the reconstruction and detection accuracy of agricultural images has not been systematically studied. Similarly, while deep learning-based object detection algorithms excel at identifying and locating objects within images, their efficacy in detecting and classifying flowers at different growth stages, particularly roses, remains underexplored.

Integrating these advanced technologies—UAV imaging, super-resolution reconstruction, and deep learning-based object detection—presents an opportunity to revolutionize rose growth monitoring. By addressing the resolution limitations of UAV images and developing algorithms specifically designed for rose detection, a highly accurate, efficient, and scalable greenhouse flower management system can be realized. This study proposes a novel approach that combines state-of-the-art super-resolution reconstruction techniques with a tailored object detection model for rose identification. This comprehensive method addresses both image resolution and detection challenges, providing a streamlined approach for precision floriculture. Our approach aims to overcome existing limitations in UAV-based agricultural monitoring and provides a comprehensive solution for precision floriculture. By integrating deep learning and object detection networks, our method enhances the accuracy of rose growth detection in low-resolution images. This method employs a Mamba-based super-resolution reconstruction algorithm in conjunction with an improved YOLOv10 model, applied to agriculture for the first time. Furthermore, this study utilizes high-resolution rose datasets captured by UAVs to train an enhanced ROSE-YOLO model, evaluating it on both original and reconstructed image sets. The analysis systematically examines how image reconstruction methods enhance detection models, highlighting their combined impact on achieving scalable and accurate rose monitoring in large-scale greenhouses. Through comparative analysis and quantitative evaluation, this research further explores the impact of different scaling factors on image reconstruction and detection accuracy and validates the proposed method's effectiveness through model visualization. We aim to push the boundaries of automated rose growth assessment.

The structure of this paper is as follows: Section 2 introduces the study site, data collection, preprocessing methods, and the selected SRR and object detection algorithms, including Mamba-based SRR techniques and the proposed ROSE-YOLO model. Section 3 outlines the implementation details, dataset preparation, and experimental results, providing a comparative evaluation of different SRR algorithms and various metrics used. Section 4 discusses the impact of scaling factors and different detection models on detection accuracy, incorporating floricultural survey data from the past five years. Finally, Section 5 summarizes the key findings and conclusions of this study.

2. Methodology

Fig. 1 presents a comprehensive workflow for analyzing aerial images of greenhouse roses using advanced computer vision techniques. The process is divided into four main stages: data collection and preprocessing, super-resolution reconstruction, object detection, and quantification.

In the data collection and preprocessing phase, an Unmanned Aerial Vehicle (UAV) captured high-definition aerial images. These images were then processed using the Scale Invariant Feature Transform (SIFT) algorithm to create a panoramic view, resulting in a rose distribution map.

The super-resolution reconstruction stage began by cropping the rose map into 512 × 512 pixel high-resolution (HR) images. These HR images underwent degradation to create 128 × 128 pixel low-resolution (LR) images. The LR images were then processed through various SRR algorithms, including SRCNN, EDSR, RCAN, Real-ESRGAN, SwinIR, and the novel MambaIR. This step aimed to enhance the resolution of the LR images back to 512 × 512 pixels. Data augmentation techniques were also applied to increase the dataset's diversity.

The object detection phase utilized both the original HR images and the SRR-enhanced images. The HR images were split into training and test sets to develop object detection models. Several detection algorithms were employed, including SSD, Faster RCNN, YOLO series, and the newly proposed ROSE-YOLO. These models were then applied to the SRR images to generate prediction outputs.

The final quantification stage involved visualization and analysis of the results. Grad-CAM techniques were used to create heatmaps that highlight regions of interest in the detected objects. The analysis focused on three key aspects: the influence of SRR magnification factors, the effect of different object detection models on SRR performance, and comparative analysis of global flower survey methods.

For a comprehensive discussion of the methodology, please refer to Sections 2.1 to 2.4.

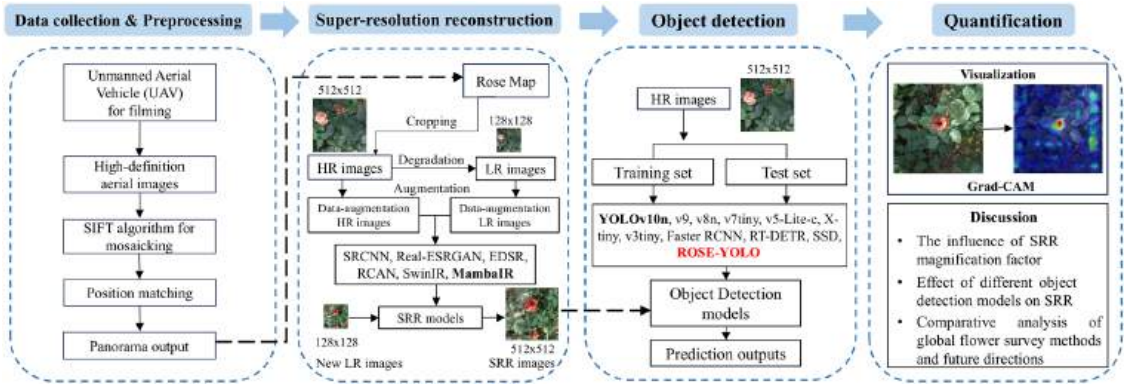

Fig. 1. Workflow of the proposed approach for the monitoring of greenhouse roses.

2.1. Survey site

The data for this study were collected from Tonghai County, Yuxi City, Yunnan Province, China, as depicted in Fig. 2(a)-(c). This region is a major producer of fresh-cut flowers, benefiting from both cost advantages and favorable climatic conditions [26]. Situated on a low-latitude plateau, Tonghai enjoys prolonged sunshine hours and high UV intensity during the winter months, which are conducive to flower growth. Additionally, the warm, humid climate and fertile soil create an optimal environment for floriculture. These natural conditions not only promote rapid growth and high yields but also ensure superior flower quality, making Tonghai a representative survey area of smart rose growth monitoring.

2.2. Data collection and preprocessing

For data collection, the consumer-grade DJI Mini 3 UAV was selected due to its excellent stability, high-resolution imaging capabilities, and extended flight time of up to 38 min [27]. The UAV's camera was configured to provide a field of view (FOV) of 90°, and it was operated at an altitude of 3 m. To ensure comprehensive coverage of all rose cultivation areas within the greenhouse, the UAV followed a serpentine flight path, which guaranteed image overlap and minimized the risk of missing any sections. Special care was taken to avoid obstructions from greenhouse structures, such as supports and shade nets, during the flight. The entire data collection process was completed in approximately 30 min.

For data preprocessing, the UAV captured high-resolution images with a resolution of 3840 × 2160 pixels, which were processed to create a panoramic view of the entire rose-growing area. The Scale-Invariant Feature Transform (SIFT) algorithm [28] was employed to integrate the captured video frames. This process involved identifying overlapping regions between consecutive frames, aligning common features, and synthesizing them into a complete panorama of the rose trellis, as illustrated in Fig. 2(d)-(g). Subsequently, the growth stages of the roses were manually annotated on the panorama using a labeling tool. The roses were classified into two stages: "rose_small" for buds with petals parallel to the central axis, and "rose_large" for flowers with petals extending beyond this point [29], as shown in Fig. 2(h). These annotated regions were then cropped into high-resolution image segments, each measuring 512 × 512 pixels.

2.3. Deep learning-based super-resolution reconstruction

The rapid advancements in deep learning have, in recent years, led to the development of various networks for image SRR. These networks aim to enhance LR images into clear, high-quality representations. Typically, SRR models are trained using datasets of paired low-resolution and HR images [30]. These pairs are created by degrading HR images with specific models to simulate the low-resolution conditions commonly encountered in practical applications.

The training of SRR models focuses on optimizing performance by minimizing a loss function that quantifies the difference between the generated Super-Resolution (SR) image and the actual HR image. Through iterative training, the model progressively learns to recover complex details characteristic of HR images, thereby enhancing the quality of the reconstructed outputs. To ensure the SRR model's efficacy, its network parameters must be meticulously calibrated to accurately capture the intrinsic relationships between LR and HR images [31]. Once trained, the SRR model can be applied to new LR images, reconstructing them into SR images and evaluating their quality and fidelity against the original HR images.

2.3.1. Network architectures

Deep learning-based SRR models have undergone significant advancements, leading to the development of various network architectures tailored to address specific challenges in image reconstruction. These approaches include Convolutional Neural Network (CNN)-based models, Generative Adversarial Network (GAN)-based models, Visual Transformer models, and more recently, Mamba-based models.

The Super-Resolution Convolutional Neural Network (SRCNN) [32] marked the first application of CNNs to super-resolution tasks. Subsequent models, such as VDSR [33], RED [34], and EDSR [35,36], built upon SRCNN by expanding the receptive field and incorporating residual learning to enhance performance. Models like RCAN [37] and VapSR [38] further improved detail preservation by integrating attention mechanisms. GAN-based models, including SRGAN [39] and ESRGAN [40], focused on improving the perceptual quality of the reconstructed images. The introduction of Visual Transformers (ViT) [41] spurred the development of models such as SwinIR [42] and ESRT [43], which are adept at capturing long-range dependencies but often come with a high computational cost. Recently, Mamba-based models have emerged as a promising alternative, with architectures like MambaIR [20] and MMA [44] enhancing computational efficiency and maintaining high performance through selectively structured state-space models [45].

This study evaluated six deep learning-based SRR architectures alongside the traditional bicubic interpolation model [46]. As depicted in Fig. 3, these models represent the evolution of SRR methodologies, each offering advancements in detail preservation, computational efficiency, and the handling of complex image structures. SRCNN laid the foundation by directly learning the mapping from LR to HR images through a simple three-layer network. EDSR enhanced this by simplifying the residual unit through the removal of the Batch Normalization (BN) layer in the ResBlock, thereby reducing complexity and parameter count while enabling deeper network structures capable of extracting more detailed features for superior performance. RCAN introduced a channel attention mechanism, which dynamically adjusts channel importance, allowing the network to prioritize high-frequency information critical for detail restoration. Real-ESRGAN [47] advanced upon

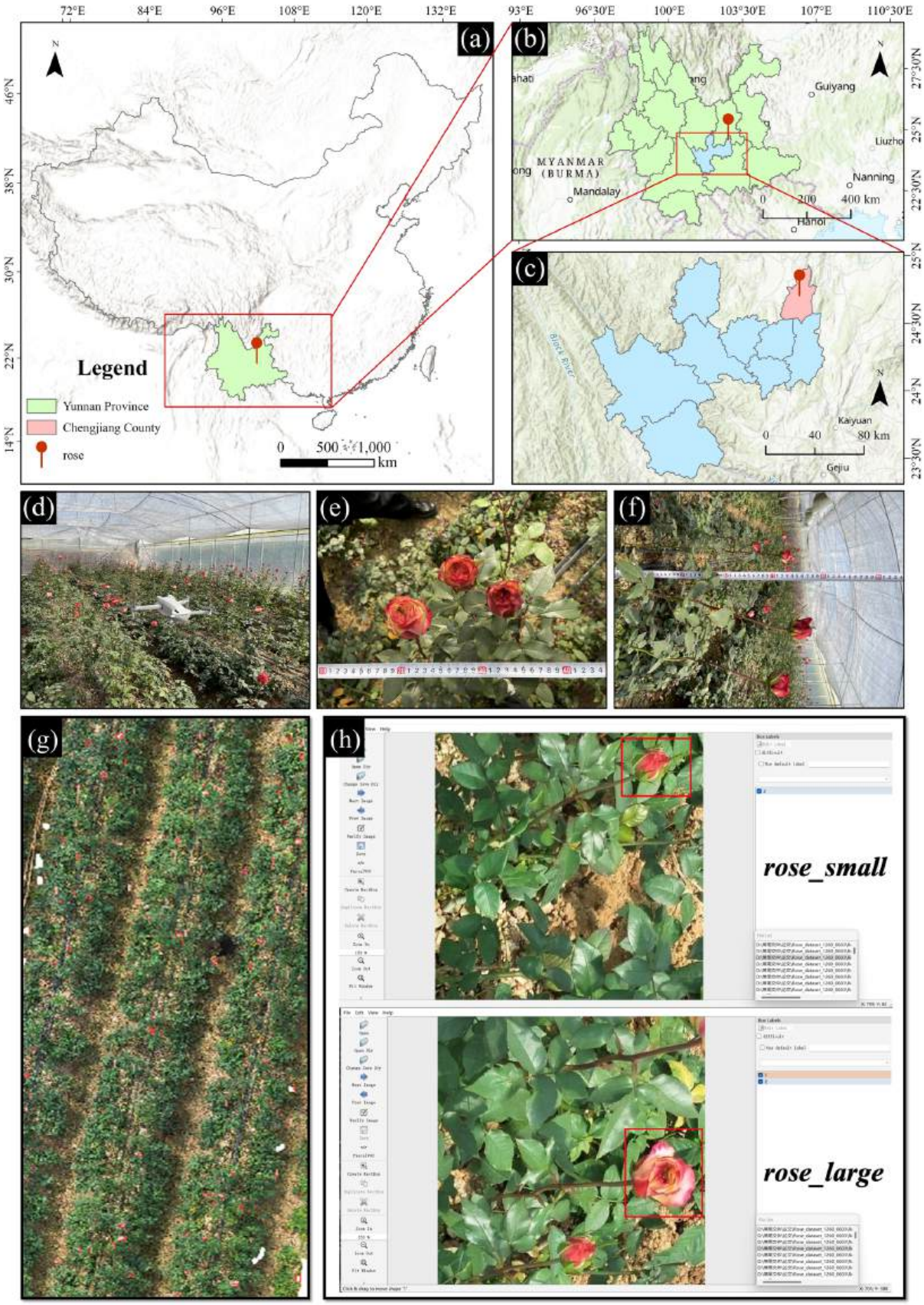

Fig. 2. (a) Location of Yunnan Province within China's administrative boundaries; (b) Administrative divisions of Yuxi City, Yunnan; (c) Map showing Tonghai County in Yuxi City; (d) Schematic of drone aerial photography techniques; (e) Sample size measurement of roses; (f) Sample view of roses; (g) Example of a rose panorama; (h) The label interface using LabelImg software.

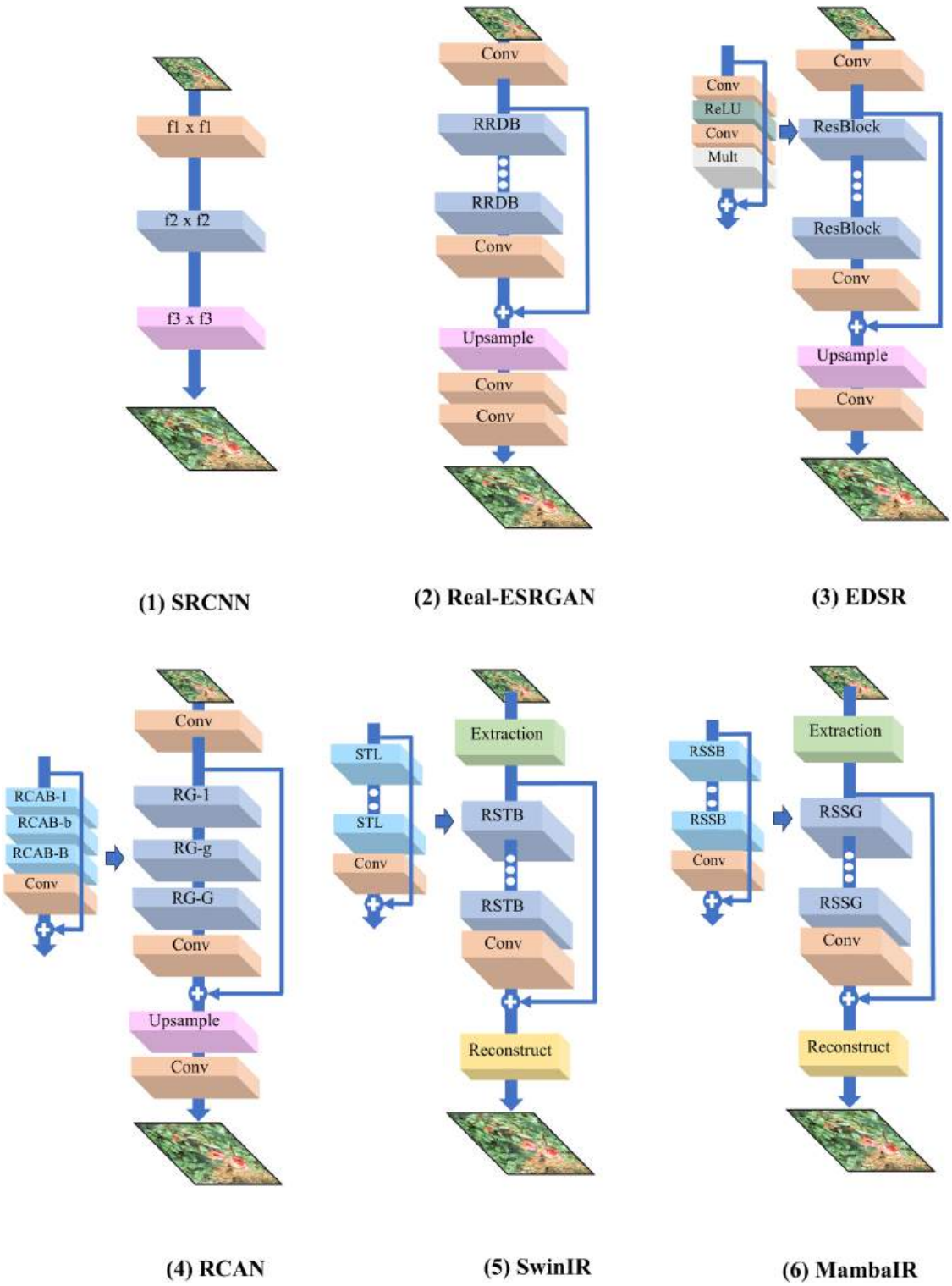

Fig. 3. Architectures of various SRR models based on different design strategies (Acronyms used in Fig. 3 are in the Supporting Information.).

ESRGAN by adopting a more sophisticated degradation model, enhancing adversarial training stability and generating HR images with richer color features and fine details. SwinIR combined the strengths of CNNs and Transformers, utilizing CNNs for efficient local feature processing and Transformers for modeling long-range dependencies, achieving high-quality outputs with fewer parameters. This model effectively mitigates noise, restores high-frequency details, reduces blurring artifacts, and produces sharper edges and more natural textures.

MambaIR represents a notable advancement by incorporating state-space modeling into image restoration. This approach addresses the local enhancement and channel-focusing challenges faced by traditional Mamba models, particularly in low-level vision tasks. With its capability to manage global receptive fields and maintain robust long-distance dependency modeling, MambaIR serves as a promising backbone for high-performance image restoration.

2.3.2. Network training and optimization

During the training of SRR models, both LR images and their corresponding HR images are used as input to the model. The model's parameters are optimized through gradient descent and backpropagation to enhance its ability to accurately reconstruct HR images. This optimization process aims to minimize a loss function, which quantifies the difference between the predicted HR image and the actual HR image in the training dataset [48].

For CNN-based models such as SRCNN, EDSR, and RCAN, the L1 loss function, also known as Mean Absolute Error (MAE), is commonly employed. The L1 loss calculates the absolute difference between the predicted and actual values, providing robustness in the presence of noisy data. The formula for L1 loss is:

$$L1 = \frac{1}{N}\sum_{i=1}^{N}|y_i - \widehat{y}_i| \tag{1}$$

where N represents the number of samples, y_i is the actual value, and $\widehat{y}_i$ is the predicted value. In contrast, GAN-based models like Real-ESRGAN utilize a combination of perceptual loss and adversarial loss for optimization. Perceptual loss evaluates image quality by comparing differences in the feature space, typically using a pre-trained convolutional neural network (CNN). The formula for perceptual loss is:

$$\text{Perceptual} = \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{M}\|\phi_j(y_i) - \phi_j(\widehat{y}_i)\|^2 \tag{2}$$

where M is the number of layers in the network, and ϕ_j denotes the features extracted from the j-th layer of the pre-trained network.

Adversarial loss is employed in GANs to ensure the generated images are indistinguishable from real images. The adversarial loss functions for the generator and discriminator are given by:

For the generator:

$$\text{Adversarial}_{\text{gen}} = -\frac{1}{N}\sum_{i=1}^{N}\log(D(\widehat{y}_i)) \tag{3}$$

For the discriminator:

$$\text{Adversarial}_{\text{disc}} = -\frac{1}{N}\sum_{i=1}^{N}[\log(D(y_i)) + \log(1 - D(\widehat{y}_i))] \tag{4}$$

where D represents the discriminator, which outputs a probability indicating whether an image is real.

Transformer-based models like SwinIR typically use the L1 loss function for optimization as well. Mamba-based models, such as MambaIR, utilize both L1 loss and Charbonnier loss, a smooth variant of L1 loss that is particularly effective in image processing tasks. The Charbonnier loss function is defined as:

$$\text{Charbonnier} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{(y_i - \widehat{y}_i)^2 + \epsilon^2} \tag{5}$$

where ϵ is a small constant, typically 10^{-3}, to prevent gradient from becoming zero. Charbonnier loss is favored for its stability in handling small errors, thereby preventing gradient explosion during training.

2.3.3. Evaluation metrics of SRR

To evaluate the effectiveness of SRR, this study employed two widely recognized metrics: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Metric (SSIM) [15]. These full-reference metrics provide quantitative assessments of image quality, enabling precise comparisons and interpretations of reconstructed images.

PSNR is a commonly used metric for assessing the similarity between a reconstructed image and its ground truth counterpart. It is defined as the ratio between the maximum possible pixel value and the Mean Squared Error (MSE) between the images. The PSNR formula is given by:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(I_{SR}^{i} - I_{HR}^{i}\right)^2 \tag{6}$$

$$PSNR = 10\cdot\log_{10}\left(\frac{MAX_I^2}{MSE}\right) \tag{7}$$

In these equations, n represents the number of pixels, $I_{HR}^{i}$is the high-resolution image, and I_{SR}^{i} represents the reconstructed image. The maximum pixel value (MAX_I) is typically 255 for 8-bit images. Higher PSNR values indicate better image quality, with typical ranges between 20 and 40 dB [49].

SSIM is a perceptual metric that evaluates image quality based on the human visual system (HVS). It measures the degradation of structural information by comparing luminance, contrast, and structural details between the reconstructed and original images. The SSIM formula is given by:

$$SSIM(x,y) = \frac{\left(2\mu_x\mu_y + K_1\right)\left(2\sigma_{xy} + K_2\right)}{\left(\mu_x^2 + \mu_y^2 + K_1\right)\left(\sigma_x^2 + \sigma_y^2 + K_2\right)} \tag{8}$$

where μ_x and μ_y are the mean luminance values of the images, σ_x and σ_y are their standard deviations, and σ_{xy} denotes the covariance between images X and Y. Constants K_1 and K_2 are used to stabilize the division. SSIM values range from 0 to 1, with higher values indicating greater structural similarity and overall image quality [50].

2.4. Object detection

2.4.1. Object detection network architectures

Deep learning-based target detection methods can be broadly categorized into two main types: two-stage and single-stage detection methods. Two-stage methods, such as the RCNN family [51–53], first generate candidate regions that are then classified and localized. In contrast, single-stage methods, including the YOLO family [54–56] and SSD [57], directly predict the location and class of targets without generating candidate regions. The YOLO family has gained popularity due to its efficiency, accuracy, and lightweight design. The original YOLO model, introduced by Redmon et al. in 2016, has evolved through multiple iterations, culminating in the YOLOv10 model [58], which significantly advances both accuracy and speed. A key innovation of YOLOv10 is its adoption of a non-maximal suppression (NMS)-free training strategy, replacing the traditional NMS step. By integrating one-to-many and one-to-one allocation methods through a dual-label allocation mechanism, YOLOv10 achieves consistency between training and inference phases, thereby reducing inference time significantly. Moreover, YOLOv10 optimizes various components of the model to enhance performance while minimizing computational costs, making it well-suited for object detection tasks that require high speed and precision.

Despite these advancements, YOLOv10 encounters challenges when applied to UAV-collected images of roses, which often feature complex backgrounds and high dynamic range [59]. To address these challenges, this study introduced the ROSE-YOLO detection model, which was built upon YOLOv10 by incorporating three key enhancements, as shown in Fig. 4: 1) Backbone Network Enhancement: The C2f structure in the backbone network is improved by replacing the Bottleneck module with the C2f-Faster module [60]. This modification enhances the model's ability to handle diverse rose datasets and generates feature maps with richer semantic information. 2) Neck Network Improvement: In the neck network, the C3CFBlockNet module [31] is employed to capture contextual information more effectively, thereby reducing the impact of occlusions caused by branches and leaves on rose detection. 3) Head Network Augmentation: The head network incorporates the BiFormer two-tier routing attention mechanism [61], which dynamically adjusts

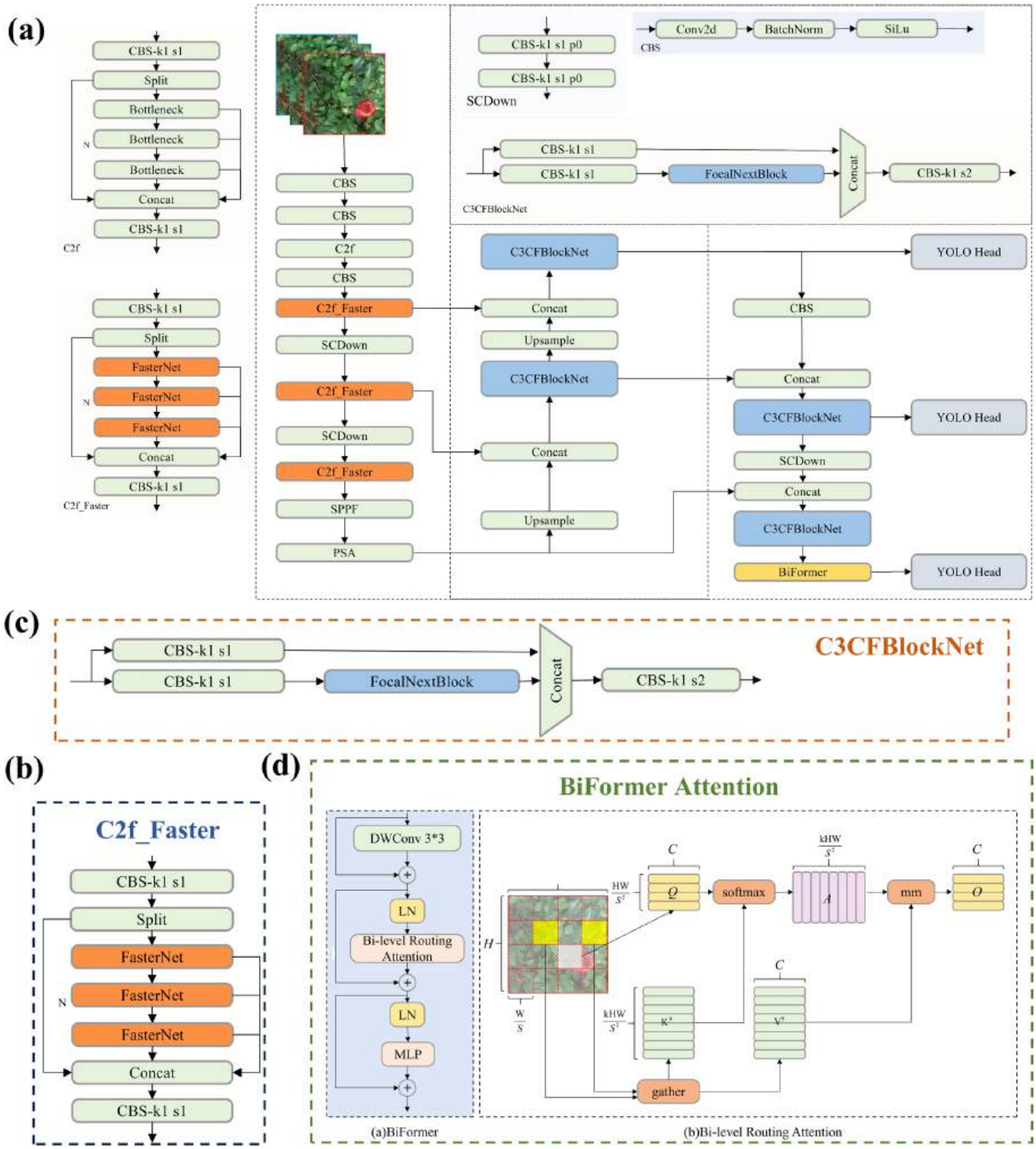

Fig. 4. (**a**) Enhanced modules and network structure of the ROSE-YOLO model; (**b**) Architecture network of C2f-Faster; (**c**) Architecture network of C3CFBlockNet; (**d**) Architecture of the BiFormer Attention mechanism. (Acronyms used in Fig. 4 are in the Supporting Information.).

feature importance based on target characteristics. This mechanism has demonstrated effectiveness in improving object detection accuracy by focusing on critical features in complex backgrounds. It enhances the model's focus on rose targets by increasing the weight assigned to roses in the feature maps.

2.4.2. Network training and optimization

The ROSE-YOLO model, a refined version of the YOLOv10 architecture, is specifically designed to classify and localize roses with high accuracy. During training, the model first extracts crucial features from input images using its backbone network. These features are then processed by the detection head to predict bounding boxes, assign class labels, and compute confidence scores. The proposed ROSE-YOLO model retains the YOLOv10 loss function, which combines Binary Cross-Entropy (BCE) for classification, emphasizing hard-to-classify samples in complex and multi-textured environments, and a

combination of Distribution Focal Loss (DFL) and Complete Intersection over Union (CIoU) for regression. This design enhances the model's ability to accurately identify and localize rose targets, particularly in challenging scenarios involving small objects or occlusions. Its well-validated design is particularly suited to the complex and dynamic detection tasks addressed in this study.

This loss function is divided into two core components: classification loss and regression loss. The classification loss is calculated using Binary Cross-Entropy (BCE), which assesses the variance between the predicted probabilities and the true labels. BCE is mathematically defined as follows:

$$L_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}[y_i\log(p_i) + (1-y_i)\log(1-p_i)] \tag{9}$$

where N denotes the total number of predictions, y_i represents the actual label of the i-th prediction, and p_i is the predicted probability. This approach yields a confidence score for each class, with the highest score indicating the confidence level for the corresponding anchor box, effectively distinguishing roses from the background.

For regression loss, the model integrates Distribution Focal Loss (DFL) with Complete Intersection over Union (CIoU) loss. CIoU is particularly effective because it takes into account several factors between the predicted and actual bounding boxes, such as the alignment of center points, diagonal distances, and aspect ratio uniformity. The CIoU loss is expressed in Eqs. (10) to (13):

$$CIoU = IoU - \left(\frac{\rho^2(b, b^{gt})}{c^2} + \alpha v\right) \tag{10}$$

$$Loss_{CIOU} = 1 - CIOU \tag{11}$$

$$\alpha = \frac{v}{(1 - IOU) + v} \tag{12}$$

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w^{gt}}{h^{gt}} - \arctan\frac{w}{h}\right)^2 \tag{13}$$

where, IoU denotes the Intersection over Union between the predicted and ground truth boxes, b and b^{gt} are the center point coordinates of the predicted and ground truth boxes, ρ^2 measures the Euclidean distance between these center points, c represents the diagonal length of the smallest box that encloses both predicted and ground truth boxes, and v measures the aspect ratio loss, which quantifies the alignment between the width and height of the predicted and ground truth boxes. By addressing these factors, the CIoU loss enhances the model's object localization capabilities. The overall loss for the ROSE-YOLO model is calculated as a weighted sum of the BCE, DFL, and CIoU losses, as shown in Eq. (14):

$$L_{Total} = \lambda_1 L_{BCE} + \lambda_2 L_{DFL} + \lambda_3 L_{EIoU} \tag{14}$$

where λ_1, λ_2, and λ_3 are coefficients that balance the contributions of BCE, DFL, and CIoU losses, respectively. The model is optimized using the Adam optimizer, which improves the stability and convergence of the training process. Consequently, ROSE-YOLO demonstrates high accuracy and robustness when processing complex and diverse datasets. This combination of standard and advanced loss components ensures that the model excels in detecting intricate rose feature.

2.4.3. Evaluation metrics for object detection

Evaluating a target detection model involves a comprehensive analysis using several key metrics. Among these, accuracy indicates the proportion of true positive predictions among all predicted positive instances. On the other hand, recall focuses on the fraction of correctly predicted positive instances relative to all actual positive instances. Another essential metric is the mAP, which assesses the performance of bounding boxes by comparing detected boxes with ground truth boxes, assigning them a score based on their accuracy [62,63]. The relevant formulas are as follows:

$$Precision = \frac{TP}{TP + FP} \tag{15}$$

$$Recall = \frac{TP}{TP + FN} \tag{16}$$

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}} \tag{17}$$

$$AP(y, y^*) = \frac{1}{N}\sum_{c} area(Pr) \tag{18}$$

$$mAP = \frac{1}{n}\sum_{i=1}^{n} AP_i \tag{19}$$

In these equations, True Positive (TP), False Negative (FN), and False Positive (FP) are critical metrics for evaluating the model's performance. TP refers to the number of samples correctly identified by the model as having flowered, FN refers to the number of samples that were actually flowered but were misidentified as not having flowered, and FP refers to the number of samples incorrectly identified as having flowered. A higher precision value indicates a lower false positive rate, while a higher recall value signifies that the model successfully captures a greater number of true positives. Intersection over Union (IoU) is another essential metric, defined as the ratio of the intersection area to the union area between the predicted bounding box and the ground truth bounding box. A higher IoU value indicates more precise localization of detected targets. Average Precision (AP) measures the balance between precision and recall, reflecting the model's ability to accurately identify objects within a specific category while minimizing false positives. However, it is important to note that AP assesses performance within a single category and does not account for performance across multiple categories. In contrast, mAP provides a comprehensive assessment of the model's performance across all object classes. mAP is the mean of AP scores calculated for different categories or classes within the object detection framework. In this study, mAP@50 is used for evaluation, which refers to the mean average precision calculated at an IoU threshold of 0.5 [64].

3. Experiment and results

3.1. Dataset description

In this study, the rose object detection model was trained using a dataset collected with a UAV, consisting of 1260 images of various roses, each with a resolution of 512 × 512 pixels. To enhance the model's representational and generative capabilities, data augmentation techniques such as rotation, symmetric flipping, and mirroring were applied [65]. These augmentations increased the diversity of the dataset, improving the model's performance and robustness. Consequently, the training set was expanded fourfold, resulting in a HR image set.

For training the SRR model, a degradation model was utilized to create a LR image dataset from the HR images [66]. The degradation process is described by the following equation:

$$g = (f \otimes h)\downarrow_s^{bicubic} + \eta \tag{20}$$

where g denotes a low-resolution image, f denotes a high-resolution image, h represents a point spread function under uniform linear motion, $\otimes$ denotes a convolution operation, $\downarrow$ indicates downsampling, bicubic denotes a bicubic interpolation algorithm, and η is the additive Gaussian white noise used to simulate real-world imaging conditions.

To further enhance data diversity and training efficiency, sub-images

of size $l_{sub} \times l_{sub}$ pixels were randomly cropped from the LR images, with corresponding HR sub-images of size $sl_{sub} \times sl_{sub}$ pixels were cropped from the same locations in the HR images. These paired LR and HR sub-images were then used as training samples. After processing, the HR dataset was divided into training, validation, and test sets, accounting for 70 %, 10 %, and 20 % of the data, respectively.

3.2. Experiment setup

This study evaluated the effectiveness of six SRR models—SRCNN, EDSR, RCAN, Real-ESRGAN, SwinIR, and MambaIR—in reconstructing rose images. Initially, all six SRR networks were trained on the same dataset. The trained models were then used to reconstruct LR images, resulting in high-quality super-resolved (SR) images. The quality of these SR images was assessed using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) metrics. Subsequently, the proposed ROSE-YOLO model was trained on the training and validation sets of the original HR rose dataset. The model's performance was evaluated across multiple datasets, including HR, Bicubic, and SR test sets. Detection performance was measured using precision, recall, and mAP metrics. Finally, Grad-CAM heatmaps were employed to visualize and analyze the model's predictions, offering insights into the impact of different SRR algorithms on target detection. This approach enhanced the transparency and interpretability of the rose detection model.

3.3. Analysis of the super-resolution reconstruction

To enhance the resolution of rose images, several networks utilizing SRR techniques were implemented, and a comprehensive performance analysis was conducted. Table 1 compares the PSNR and SSIM values of images reconstructed by seven different methods. The results indicate that deep learning-based SRR techniques generally outperform the traditional Bicubic method, achieving higher PSNR and SSIM values, signifying superior image reconstruction capabilities. However, the Real-ESRGAN method is an exception, producing relatively lower PSNR and SSIM values. This anomaly is attributed to Real-ESRGAN's unique loss function, which incorporates perceptual loss, prioritizing the reconstruction of high-frequency details and generating finer visual features [67]. In contrast, other models typically rely on pixel-based loss functions [68]. Among the tested networks, MambaIR performed the best, achieving PSNR and SSIM values of 28.34 dB and 70.07 %, respectively, highlighting the potential of state-space models with selection mechanisms and hardware-aware architectures.

Fig. 5 presents a comparison between the original HR image and those reconstructed using various methods. The Bicubic interpolation method struggles to accurately reproduce the fine line textures of the HR image, resulting in blurred details. In contrast, deep learning methods generate images with sharper edges and more accurate textures, closely resembling the original HR image and significantly outperforming the Bicubic approach. However, CNN-based models still face challenges in recovering high-frequency details, which often leads to overly smooth textures in areas with dense details [69,70]. While GAN-based methods produce more realistic images, their texture reproduction still deviates from the reference image, failing to capture all the fine details. Transformer-based models improve upon clarity and texture detail compared to both CNN and GAN approaches, yet they still do not match the performance of the Mamba-based approach. MambaIR, leveraging its broad global receptive field, excels in resolving fine details and preserving textures, resulting in a more accurate reproduction of the rose's growth environment. This superior performance is attributed to MambaIR's ability to capture both local and long-range dependencies effectively, which is crucial for maintaining high-fidelity image features. While deep learning techniques significantly enhance image clarity and detail restoration, they also introduce challenges such as excessive edge smoothing, reflecting the inherent trade-off between image clarity and realism in SRR tasks. Despite these challenges, the advancements made by these models in rose image reconstruction represent a substantial improvement over traditional methods, highlighting the potential of deep learning-based SRR models in achieving high-quality image reconstruction.

3.4. Explainable object detection result with Grad-CAM

The performance of the Bicubic method and six deep learning-based SRR techniques was evaluated using the ROSE-YOLO model across various test sets. Table 2 summarizes the detection results for HR, Bicubic, and SRR test sets derived from each algorithm. Notably, the HR test set achieved the highest mAP of 95.3 %, underscoring ROSE-YOLO's exceptional accuracy in rose detection [29]. Among the SRR methods, MambaIR demonstrated the best performance, with a mAP of 94.1 %, surpassing the Bicubic method by 7.5 % and falling only 1 % short of the HR test set. This indicates that SRR significantly enhances image quality and, consequently, detection accuracy. Moreover, it suggests that accurate object detection can be achieved with cost-effective, low-resolution, and portable equipment, thereby reducing data acquisition requirements.

Conversely, the Real-ESRGAN test set exhibited the lowest performance, with a mAP of 91.2 %, likely due to artifacts and unrealistic textures introduced during the reconstruction process, which adversely affected detection accuracy. The other SRR test sets yielded mAP values close to one another and significantly exceeded those of the Bicubic method, with an average improvement of approximately 6.5 %. As illustrated in Fig. 6(a) and (b), improved SRR metrics correlate with enhanced image quality, leading to higher mAP and confidence scores.

In object detection, the confidence score ranges from 0 to 1 and represents the model's certainty about the presence of an object within the detection region, with higher scores indicating greater confidence [71]. Overall, these results highlight the effectiveness of deep learning-based SRR techniques in improving the performance of object detection models like ROSE-YOLO, particularly in scenarios where high-resolution image acquisition is not feasible. The use of SRR not only enhances the detection capabilities of models but also supports more accessible and scalable solutions for real-world applications in agricultural monitoring and beyond.

To visually assess the impact of various super-resolution reconstruction algorithms on the ROSE-YOLO rose detection model, Gradient-weighted Class Activation Mapping (Grad-CAM) was employed [72]. Grad-CAM is a computer vision technique that enhances the interpretability of neural networks by highlighting the regions in an input image that most influence the network's predictions. It achieves this by computing weights for each channel of the output feature map from the last convolutional layer and using these weights to generate a class activation map (CAM), i.e., a heatmap [73]. This map highlights the key areas of the image, thereby elucidating the network's decision-making process.

Fig. 6(c) and (d) display heatmaps overlaid with the target detection results. The Grad-CAM results indicate that activation regions are distributed globally across the image rather than being confined strictly

Table 1
Evaluation metrics of various SRR methods on the LR test set with the magnification factor 4.

Metrics	Bicubic	Real-ESRGAN	RCAN	SRCNN	EDSR	SwinIR	MambaIR
PSNR (dB)	27.08	24.82	27.89	28.07	28.30	28.30	**28.34**
SSIM (%)	71.72	66.20	76.13	75.87	76.97	77.07	**77.07**

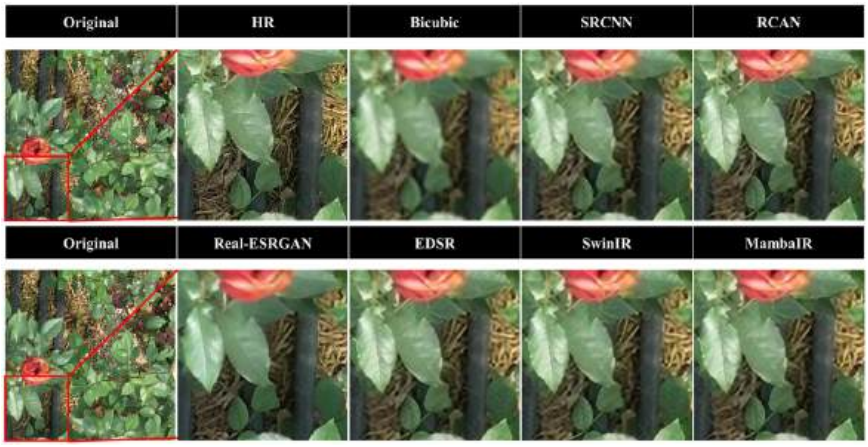

Fig. 5. Comparison of the visual effects of reconstructed images based on the seven reconstructed methods.

Table 2
Evaluation metrics of detection results of the testset reconstructed by different SRR algorithms.

Metrics (%)	HR	Bicubic	Real-ESRGAN	RCAN	SRCNN	EDSR	SwinIR	MambaIR
mAP@50	95.3	86.9	91.2	92.6	93.3	94.1	94.1	**94.4**

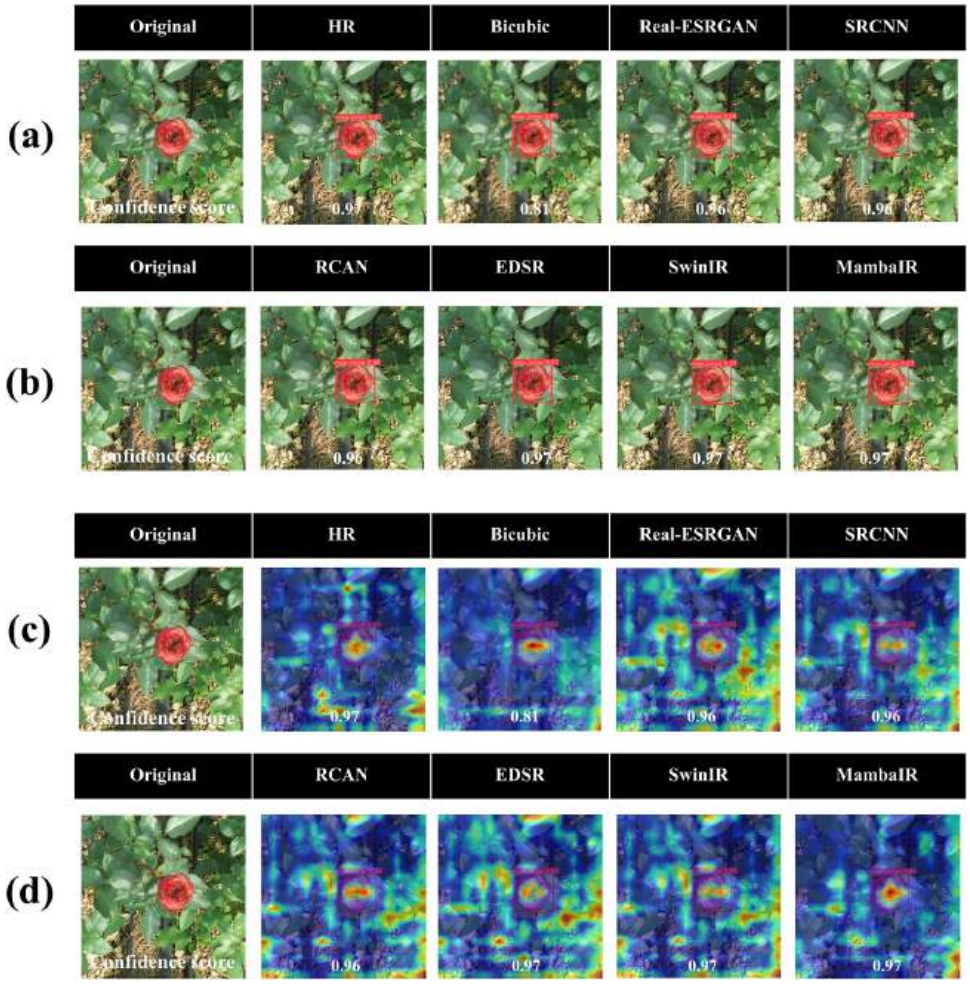

Fig. 6. Qualitative comparison of rose detection results of various test sets using ROSE-YOLO.

to the detected bounding boxes. The brighter areas on the heatmap correspond to regions where the model strongly predicts the presence of a rose [74]. For the ROSE-YOLO model, the HR test image achieved the highest confidence score of 0.97, with highlighted areas concentrating on the degree of petal opening and bending. Conversely, the Bicubic interpolation method produced the lowest confidence score of 0.81, with the red highlights in the heatmap deviating from the actual rose, indicating substantial discrepancies in the algorithm's reconstruction capabilities.

Among the deep learning-based SRR algorithms, confidence scores ranged from 0.96 to 0.97, demonstrating their superiority over traditional methods. However, variations in the heatmaps reveal differences in how these algorithms restore image details and overall structure. For example, the Real-ESRGAN algorithm's heatmap exhibited multiple red-highlighted regions, suggesting the introduction of unrealistic artifacts during image reconstruction, leading to false detections. This issue arises because, while Real-ESRGAN enhances global visual perception, it struggles with maintaining local detail fidelity, causing the model's attention to drift toward background elements like leaves and dirt.

Similar issues were observed with CNN-based algorithms such as RCAN and EDSR. Despite their high confidence scores and the heatmaps' emphasis on rose petals, the inherently local nature of CNN operations limits their ability to capture global contextual information. Consequently, some attention is diverted to the surrounding background, which can affect overall detection performance. Interestingly, the SRCNN algorithm, as an earlier development in super-resolution reconstruction, showed different characteristics. Its heatmap had minimal red highlights, indicating a strong ability to suppress background noise and maintain a balance between global and local information. Although SRCNN has limitations in enhancing image resolution, its stable performance has provided a solid foundation for the advancement of subsequent super-resolution algorithms.

The Transformer-based SwinIR algorithm also demonstrated stable performance in the ROSE-YOLO model, indicating an advantage over CNNs in image reconstruction tasks. However, like SRCNN, SwinIR's attention was partly directed to nearby leaves, with less focus on the rose itself, particularly around the edges and curves of the petals. This suggests a limitation in capturing the overall shape.

Finally, the MambaIR-based algorithm achieved a confidence score of 0.97, matching that of the HR test image, underscoring its superiority in rose reconstruction. MambaIR's broader receptive field effectively leverages the repetitive priors of image blocks, significantly enhancing detail and texture retention. The red highlights in its heatmap closely aligned with the rose, even more accurately than the HR image. This could be attributed to fine-tuning of illumination and color during super-resolution reconstruction, optimizing the image and enhancing the model's ability to discern distinct visual patterns [19]. Additionally, compared to the HR test image, the overall red highlights were slightly reduced in MambaIR's heatmap, suggesting that MambaIR effectively minimizes irrelevant details during reconstruction, allowing the model to focus more precisely on the key features. This indicates that MambaIR balances global and local features, presenting a promising direction for future research to further refine super-resolution algorithms in target detection tasks.

3.5. Ablation experiment for ROSE-YOLO

Detecting trellis roses captured by UAVs poses substantial challenges due to the complex environmental backgrounds, diverse rose shapes, and significant individual differences among the roses. The quality of the images further complicates the detection process. To address these challenges, the YOLOv10 model was enhanced to improve its rose detection capabilities, using the original YOLOv10, which achieved a mAP of 87.2 %, as the baseline. The results of these enhancements to the backbone, neck, and head networks are summarized in Table 3.

In the YOLO series, the design and selection of the backbone network are crucial for optimizing model performance. The introduction of the C2f-Faster module into the backbone of the original YOLOv10 led to the development of the YOLOv10-N1 model. The C2f-Faster module incorporates advanced convolutional techniques such as partial convolution (PConv) and pointwise convolution (PWConv), creating an efficient "T" convolutional structure. This structure significantly improves the model's ability to handle complex rose bush scenes by optimizing feature extraction and fusion while reducing unnecessary computational redundancy and overall computational complexity. These enhancements result in increased detection accuracy and reduce the number of parameters and model size, achieving precision, recall, and mAP values of 83.7 %, 82.3 %, and 90.7 %, respectively.

Building on YOLOv10-N1, the C3CFBlockNet module was introduced into the neck network, further refining the model into YOLOv10-N12. The C3CFBlockNet, which comprises the C3 module and the FocalNextBlock, substantially enhances the model's capacity to capture contextual information. The FocalNextBlock effectively captures long-range correlations between spatial features with only a slight increase in parameters and computational overhead. By integrating the FocalNextBlock, the C3 module forms a highly efficient, multi-scale C3CFBlockNet, which significantly mitigates the impact of contextual factors on rose detection. This enhancement boosts the mAP to 94.1 % while further reducing the model's size, ensuring efficient parameter utilization and improved detection performance.

Finally, the BiFormer module was incorporated into the head network, resulting in the YOLOv10-N123 (ROSE-YOLO) model, also referred to as ROSE-YOLO. The BiFormer module utilizes a bi-level Routing Attention (BRA) mechanism to dynamically and efficiently allocate computational resources. By increasing the weight of rose features in the feature map and decreasing the weight of background features, the module sharpens the model's focus on rose detection. The integration of BiFormer significantly enhances recall while also increasing precision and mAP values, culminating in optimal detection performance.

These progressive enhancements to the YOLOv10 model demonstrate a systematic approach to overcoming the inherent challenges of UAV-based rose detection, paving the way for more accurate and efficient agricultural monitoring applications.

4. Discussion

4.1. The influence of the SRR magnification factor

Previous studies have demonstrated that the reconstruction quality

Table 3
Ablation experiments of different detection networks.

model	N1: C2f_Faster	N2: C3CFBlockNet	N3: BiFormer	Precision (%)	Recall (%)	mAP@50(%)	Size (MB)
YOLOv10	×	×	×	7 7.9	77.1	87.2	5.8
YOLOv10-N1	√	×	×	83.7	82.3	90.7	5.1
YOLOv10-N12	√	√	×	8 4.9	85.7	94.1	4.9
YOLOv10-N123	√	√	√	86.7	90.5	95.3	5.5

*A check mark (√) indicates the strategy module was used and a cross (×) indicates it was not used.

of super-resolution (SR) images is significantly influenced by the magnification factor [22,75,76]. This critical parameter, which dictates the upsampling factor, determines the increase in size from an LR image to a super-resolution (SR) image, thereby reflecting the extent of resolution enhancement achieved through SRR techniques. To identify the optimal magnification for rose detection, the following experimental procedure was employed: First, images with a resolution of 512 × 512 pixels were preprocessed and designated as a HR rose dataset. This dataset was divided into training, validation, and test sets in proportions of 70 %, 10 %, and 20 %, respectively. The HR images were downsampled by a factor of four to create an LR dataset, which was subsequently used to apply the deep learning-based SRR method. The MambaIR network was chosen to examine the impact of different magnifications on SRR performance, with magnification factors set at 2, 3, 4, and 5.

Initially, the x2_SRR model was trained using the LR training set and the original HR dataset. This x2_SRR model was then applied to the x1_LR test set (128 × 128 pixels), generating the x2_SR test set (256 × 256 pixels). Similar procedures were followed to create the x3_SR (384 × 384), x4_SR (512 × 512), and x5_SR (640 × 640) test sets. Subsequently, the HR rose dataset's training and validation sets were used to train the ROSE-YOLO model, which was then applied to the SR images reconstructed at different magnifications [77,78]. The detection performance of these SRR-reconstructed images was evaluated using mAP and confidence scores. Table 4 illustrates the ROSE-YOLO model's recognition accuracy for rose images across various magnifications.

Fig. 7 presents detection results for reconstructed rose images at different magnifications. Generally, higher magnifications produce SR images with more detailed features, improving accuracy in tasks like object detection. However, this improvement comes at the cost of increased computational complexity and processing time. Specifically, as magnification increases, the gap between detection results and actual labels decreases, but computational and time requirements escalate. For instance, compared to the x3 magnification, computation times for x4 and x5 increased by 20 % and 50 %, respectively. A magnification factor of 4 offers an optimal balance between detection accuracy and computational efficiency. This finding is consistent with existing research, which suggests that magnification factors of 3x or 4x generally strike a balance between detail reconstruction and computational efficiency [79,80]. In greenhouse environments with lower altitudes, a smaller but suitable magnification factor, such as 2x, may suffice to achieve detailed monitoring without introducing unnecessary computational burdens. Conversely, in open-field scenarios where UAVs operate at higher altitudes and cover larger areas, higher magnification factors, such as 4x or even 8x, could be employed to enhance resolution and detect fine details in crops. These adjustments should be tailored to the plant characteristics and monitoring objectives, as further discussed in Section 4.3. Additionally, higher magnifications may introduce artifacts and distortions, potentially degrading recognition results, highlighting the importance of selecting an appropriate magnification factor.

4.2. Effect of different object detection models on SRR

To evaluate the performance of various object detection networks in rose detection, ten models—including Faster-RCNN, Mobile-SSD, RT-DETR, and several YOLO variants—were trained using high-resolution rose datasets collected by UAVs. Table 5 presents the detection performance of these models across three test sets: high resolution (HR), Bicubic interpolation (Bicubic), and MambaIR (super-resolution reconstruction).

Table 4
Evaluation metrics of detection results of the test set reconstructed by different magnification factors.

Metrics (%)	x1_LR	x2_SR	x3_SR	x4_SR	x5_SR
Precision	74.1	77.2	80.1	92.1	85.5
Recall	81.7	86.2	89.8	93.7	90.8
mAP@50	77.6	91.0	93.4	**94.4**	93.3

Our analysis shows that the proposed ROSE-YOLO model outperforms all baseline models across each test set. Notably, on the HR test set, ROSE-YOLO achieved a mAP that was 6.7 % higher than the second-ranked YOLOv6n, highlighting its superior capability to process high-resolution images effectively. On the Bicubic test set, although all models showed reduced performance due to the lower resolution, FML-YOLO maintained the highest mAP, indicating its robustness in handling low-resolution images. Similarly, ROSE-YOLO outperformed other models on the MambaIR test set, achieving a mAP that was 6.6 % higher than YOLOv6n, demonstrating its enhanced performance in reconstructing super-resolution images. To provide deeper insights into the model's performance, per-class evaluation metrics, including precision, recall, and F1-score for large roses and small roses, have been analyzed and are detailed in the Supporting Information. This additional evaluation highlights the effectiveness of the proposed ROSE-YOLO model across both classes, enabling a more granular understanding of its capabilities.

The results indicate that two-stage detectors, such as Faster R-CNN, do not outperform one-stage models like SSD and the YOLO series. Faster R-CNN, despite its larger model size, exhibits lower efficiency across all metrics compared to ROSE-YOLO, primarily due to differences in model architecture. As a two-stage detector, Faster R-CNN first generates candidate regions using a Region Proposal Network (RPN) and subsequently classifies and refines these regions [81]. This process entails substantial computational complexity and time overhead, rendering it less suitable for real-time detection. In contrast, one-stage detectors directly generate class probabilities and positional coordinates in a single step, leading to faster and more efficient outcomes [82]. Notably, Faster R-CNN recorded the lowest mAP on the Bicubic test set, underscoring its limitations in handling low-resolution images.

Certain YOLO variants, including YOLOv3-tiny and YOLOv5Lite-c, yielded low detection accuracy in this context. YOLOv3-tiny, a simplified version of YOLOv3, features a basic network structure and lower computational requirements, which compromise its detection capabilities [83]. Similarly, YOLOv5Lite-s, with its reduced parameter count, lacks the complexity and representational capacity needed to effectively capture and identify subtle features of roses, resulting in diminished accuracy. In contrast, ROSE-YOLO combines high accuracy and robustness with a minimal parameter count and compact model size. Its lightweight design makes it well-suited for high-performance detection tasks and enables efficient operation in resource-limited environments, expanding its potential applications. In future implementations, ROSE-YOLO could be integrated into embedded systems or mobile devices, such as UAVs or agricultural robots, for real-time crop monitoring.

Fig. 8 presents an interpretive analysis of detection results from various networks on high-resolution images using Grad-CAM. Grad-CAM heatmaps reveal the feature recognition logic of different object detection models. Notably, only YOLOv6n and ROSE-YOLO accurately highlight regions covering the target object with high confidence. In contrast, heatmaps from other models tend to focus on background noise or irrelevant features, aligning with earlier analyses. ROSE-YOLO, in particular, shows a significant advantage, achieving the highest confidence score of 0.94 on test images and displaying the most extensive highlighted regions in its heatmap.

In summary, the proposed ROSE-YOLO substantially enhances the detection accuracy of the YOLO series in rose detection, showing significant improvements in accuracy, recall, and mAP. These results indicate that ROSE-YOLO is more adept at detecting and identifying roses across a variety of scenarios, effectively minimizing false negatives and false positives, thereby improving detection reliability and efficiency. Additionally, ROSE-YOLO 's robust performance in detecting objects within complex backgrounds and across diverse flower types offers an effective solution for precision agriculture applications.

Fig. 7. Detection results of rose images from SRR testsets obtained with different magnification factors.

Table 5
The detection results of competitive networks on various testsets.

Different models	Size (MB)	HR (%)			Bicubic (%)			MambaIR (%)		
		P	R	mAP@50	P	R	mAP@50	P	R	mAP@50
Faster R-CNN	113.5	69.8	89.9	87.4	64.7	86.6	75.0	65.3	89.6	83.0
RT-DETR	18.2	76.9	81.0	86.6	66.4	**89.6**	81.4	84.1	75.5	85.6
Mobile-SSD	15.5	69.3	88.0	85.8	67.1	85.0	76.1	67.0	91.9	84.6
YOLOv3tiny	17.4	81.5	81.7	83.8	79.4	85.3	79.2	84.3	80.2	83.4
YOLOv6n	8.7	85.0	79.2	88.6	77.6	79.6	83.7	86.4	76.6	87.8
YOLOv5Lite-c	9.1	83.3	83.6	86.3	74.3	84.0	75.7	78.6	86.5	82.7
YOLOX-tiny	20.4	71.3	**90.8**	87.0	65.7	88.9	77.6	67.3	90.9	83.1
YOLOv7tiny	12.3	86.4	77.3	88.3	78.2	83.3	84.6	84.5	76.6	86.6
YOLOv8n	6.3	76.6	89.2	87.8	77.7	81.4	83.3	79.8	84.1	87.3
YOLOv9	7.9	81.7	83.9	87.2	74.5	83.0	82.6	83.3	83.4	86.7
ROSE-YOLO	5.5	**86.7**	90.5	**95.3**	**84.4**	87.3	**86.9**	**92.1**	**93.7**	**94.4**

4.3. Comparative analysis of global flower survey methods and future directions

To comprehensively compare and discuss global flower survey methods and results, this paper analyzes the application and effectiveness of various survey techniques across different regions and flower species. The goal is to provide valuable guidance and references for future research. By synthesizing insights from prior studies and the results of this research, it highlights advancements in imaging technologies and detection algorithms while contextualizing their practical applications in floriculture.

Fig. 9 illustrates representative flower survey regions and species covered over the past five years, including major flower-growing areas in Asia, Europe, and North America [84,85]. The study encompasses species such as roses, apple blossoms, and lily of the valley, all of which hold significant economic and scientific importance [86]. These species represent a diverse array of ecological and agricultural contexts, underscoring the need for adaptable survey methods. The inclusion of diverse geographic regions ensures that the analysis captures the varying environmental, economic, and technological contexts influencing flower surveys globally.

Survey algorithms vary by region due to differences in flower growth characteristics and environmental conditions. Generally, these algorithms fall into two categories: detection algorithms and segmentation

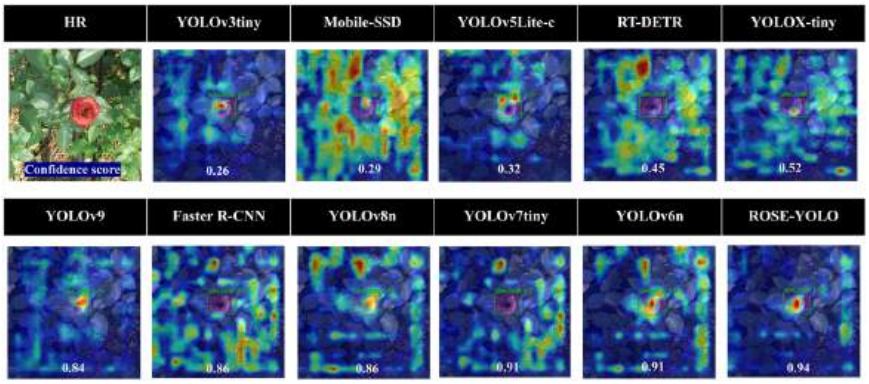

Fig. 8. Explainable detection results of different detection networks on HR image using Grad-CAM.

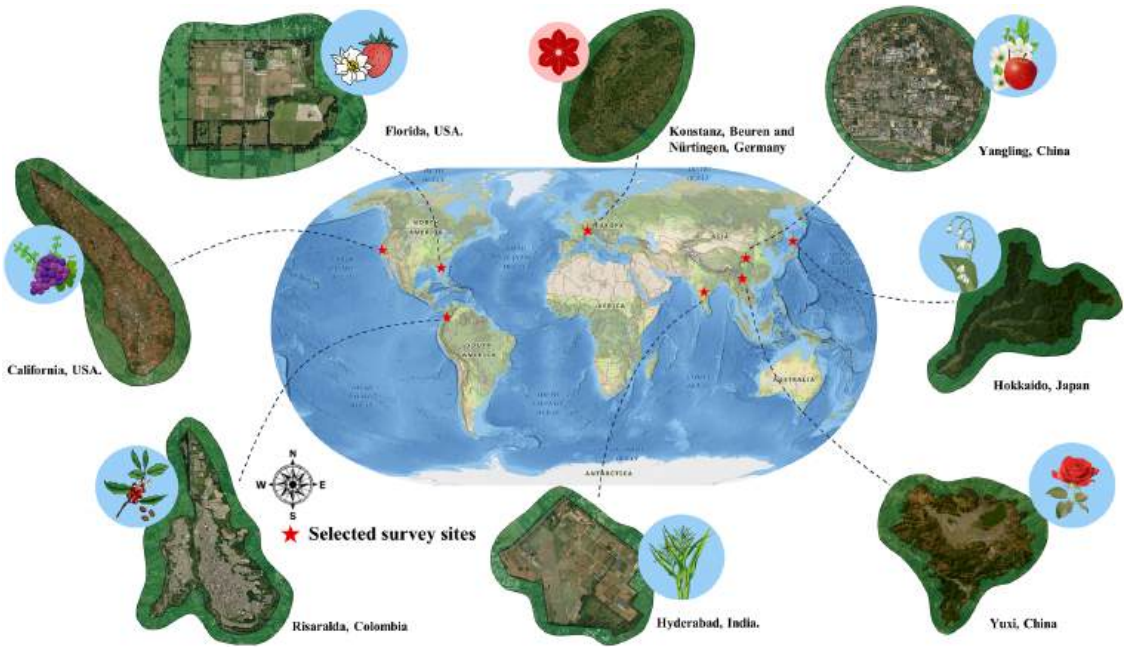

Fig. 9. Geographic distribution of studies that used techniques to monitor and detect different types of flowers.

algorithms. Detection algorithms are more suitable for larger, sparser flowers that are easily distinguishable from backgrounds, such as roses and apple blossoms [10]. These flowers occupy a significant portion of the image and have distinctive features, making them ideal candidates for single-stage detection algorithms like YOLO, which performs effectively with large datasets. In contrast, segmentation algorithms excel with smaller, denser flowers, such as grapevine inflorescences [11]. Due to their subtle morphological features and tendency to blend with the background, segmentation algorithms provide precise pixel-level categorization, allowing for detailed separation of individual flowers and accurate quantification, especially in complex backgrounds.

Image acquisition primarily relies on UAV imaging technology, which has become the standard for flower surveys due to its flexibility and extensive coverage. UAVs equipped with high-resolution cameras can efficiently capture large planting areas, providing data at lower costs and higher speeds. The adjustable flight altitude and speed of UAVs ensure the capture of clear, comprehensive images under various terrain conditions, offering high-quality input for subsequent algorithmic processing. While manual imaging has been employed in some studies, it is limited by high labor and time requirements and is susceptible to environmental factors such as lighting and weather, which can affect data consistency and reliability. Comparisons of UAV and near-ground imaging methods in strawberry surveys at the Citra Plant Science Research and Education Center in Florida revealed that image quality deteriorates when drones fly above 3 m, affecting the visibility of flowers and fruits [87]. This highlights the importance of integrating advanced image enhancement techniques to overcome these limitations.

To address the limitations of UAV-based imaging, this study introduces a super-resolution reconstruction (SRR) algorithm. The SRR algorithm enhances low-resolution (LR) images to near-high-resolution (HR) quality, reducing the reliance on high-precision imaging equipment while improving detection accuracy. This technology demonstrates significant potential in UAV image processing, particularly in scenarios where image quality is compromised by altitude or environmental factors.

Fig. 10 presents the ROSE-YOLO model's results for rose detection, with a rose distribution map generated by stitching together images using the SIFT algorithm. The red boxes highlight successfully detected roses, demonstrating the model's efficiency and accuracy in recognizing and mapping rose distribution. The results show that integrating high-resolution images from the MambaIR SRR network with the ROSE-YOLO model enables comprehensive mapping of rose detection in UAV images. This integration not only enhances detection accuracy but also significantly improves image resolution, facilitating fine-grained

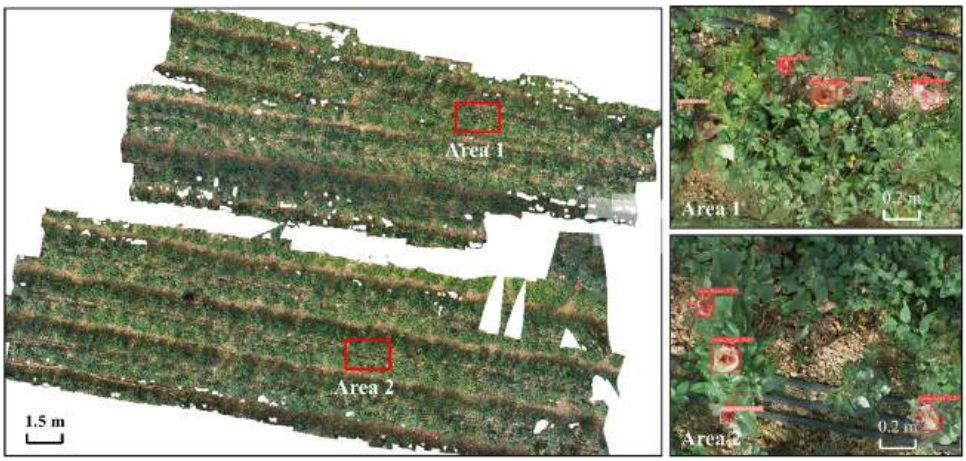

Fig. 10. Rose detection and distribution map.

analysis across extensive planting areas. These high-resolution images offer valuable insights for ecological analysis, enabling precise identification of rose growth conditions and detailed growth maps, thereby supporting future cultivation management and scientific research [88].

Looking ahead, the ROSE-YOLO model will be expanded to classify additional growth stages, such as initial bloom, full bloom, and wilting. This finer classification will provide detailed insights into growth dynamics, aiding in the optimization of floriculture strategies to enhance yield and quality [89]. Beyond classification, the integration of features extracted from reconstructed images offers the potential for predictive modeling of rose growth trends and health. By analyzing petal structures, early signs of disease, or growth rates visible in high-resolution reconstructed images, the model could be extended to predict resource requirements such as irrigation or fertilization, enabling more proactive and efficient management strategies [29].

Additionally, training the ROSE-YOLO model on various flower species will broaden its application, enabling accurate classification across multiple flower types. This advancement will significantly improve the efficiency of precision agriculture and foster the development of intelligent, data-driven agricultural practices [90]. ROSE-YOLO is poised to become a core tool in precision agriculture, advancing towards smarter, more sustainable farming practices and addressing global agricultural challenges, including increased yields, reduced resource wastage, and improved product quality.

This summary analysis highlights the strengths and limitations of current flower survey methods while demonstrating the innovative contributions of this study. By combining UAV imaging, SRR technology, and advanced detection algorithms, this research sets a benchmark for future work in intelligent agricultural practices, particularly in floriculture. The integration of these techniques provides valuable insights for the global agricultural community, paving the way for smarter and more sustainable farming solutions.

5. Conclusions

This study introduced a novel approach for monitoring rose growth in large greenhouse environments by combining UAV imaging with advanced deep-learning techniques. We applied MambaIR, an SRR algorithm based on Mamba, which significantly enhanced the quality of low-resolution UAV images. MambaIR outperformed traditional and other deep learning-based SRR methods, achieving a PSNR of 28.34 dB and an SSIM of 77.07 %. We proposed ROSE-YOLO, an improved object detection model based on YOLOv10, specifically tailored for rose detection. ROSE-YOLO achieved a mAP of 95.3 % on high-resolution images, surpassing other state-of-the-art detection models. The combination of MambaIR and ROSE-YOLO demonstrated significant synergy, achieving a 94.4 % mAP on reconstructed low-resolution images, which is only 0.9 % lower than that of the original high-resolution images. This indicates that our approach has the potential to overcome resolution limitations in UAV-based agricultural monitoring. We identified an optimal super-resolution magnification factor, balancing detail enhancement and computational efficiency. Grad-CAM visualization results show that our method effectively focuses on key rose features, outperforming other models in attention allocation and feature extraction. This study presents a scalable, efficient, and accurate method for rose growth monitoring, improving the quality of agricultural data collection and demonstrating the potential of integrating deep learning techniques into traditional remote sensing to overcome its limitations. Future work should focus on extending the model's capabilities to classify multiple growth stages and different flower species, further advancing intelligent, data-driven agriculture.

Author's contributions

F.Z. conceptualized the research plan, developed the data processing flow, and analyzed the experimental data; F.Z. and Y.C. conducted the field survey. F.Z. and K.M. proposed the idea for super-resolution and detection of rose flower. F.Z., J.W., D.X., and Z.R. created the training data for deep learning and visualized the results. Q.W., Y.L., X.S., and K. M. revised the manuscript. F.Z. implemented the deep learning method and analyzed the data. All authors discussed the results. F.Z. and Z.R. wrote the manuscript.

Ethics statement

Not applicable: This manuscript does not include human or animal research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors are grateful to Miss. Yulun Chen (Department of Environmental Science, Southwest Forestry University) for her assistance on scientific research activities in Yuxi city, China. This research was partially supported by the Japan Science and Technology Agency SPRING Program (JST SPRING), Grant Number JPMJSP2108, and the Grant-in-Aid for Scientific Research (KAKENHI) (20KK0238) provided

by the Japan Society for the Promotion of Science (JSPS).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.atech.2024.100730.

Data availability

Data will be made available on request.

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